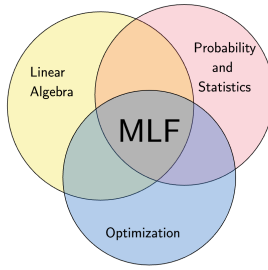


Quiz-1 | Practice Problems



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Note to Learners: [Live session](#) where these problems were discussed. Some of these problems might be on the harder side and one shouldn't be discouraged if one is not able to understand them right away.

1. Question-1

A square matrix A of shape 3×3 has eigenvalues 1, 3, 2. What are the eigenvalues of the matrix A^T ?

The characteristic polynomial of A^T is $|A^T - \lambda I|$:

$$\begin{aligned} |A^T - \lambda I| &= |A^T - \lambda I^T| \\ &= |(A - \lambda I)^T| \\ &= |A - \lambda I| \end{aligned}$$

Therefore, the characteristic polynomial of A and A^T are the same. From this, it follows that the eigenvalues of A and A^T are the same.

2. Question-2

If A is a square matrix, what can you say about the eigenvectors of A and A^T ? Are the eigenvectors of A and A^T equal?

Not necessarily.

Counter-example

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$

Finding the eigenvalues:

$$\begin{aligned}(1 - \lambda)^2 - 4 &= 0 \\ 1 - \lambda &= \pm 2 \\ \lambda &= -1, 3\end{aligned}$$

Finding the eigenvectors for $\lambda = -1$

$$\begin{aligned}(A - (-1)I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ (A + I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}\end{aligned}$$

$(1, -2)$ is an eigenvector.

Is $(1, -2)$ an eigenvector of A^T :

$$A^T = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$\begin{aligned}A^T \begin{bmatrix} 1 \\ -2 \end{bmatrix} &= \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} -7 \\ -1 \end{bmatrix}\end{aligned}$$

It isn't. Therefore, $(1, -2)$ is not an eigenvector of A^T .

Remark: A and A^T have the same eigenvalues but not necessarily the same eigenvectors.

3. Question-3

If $(1, 3, 2, 4)$ is an eigenvector of a square matrix with eigenvalue 3, what

can you say about the vector $(3, 9, 6, 12)$?

Let $u = (1, 3, 2, 4)$. Then, observe that $v = (3, 9, 6, 12) = 3u$.

$$\begin{aligned}Av &= A(3u) \\ &= 3(Au) \\ &= 3(3u) \\ &= 3v\end{aligned}$$

We see that v is an eigenvector of A with eigenvalue 3.

Remark: If u is an eigenvector of A with eigenvalue λ , then ku is also an eigenvector with the same eigenvalue λ , where $k \neq 0$.

4. Question-4

If the eigenvalues of a 2×2 matrix, A , are 2, 5 what are the eigenvalues of $A - 5I$?

If (λ, u) is an eigenpair of A :

$$Au = \lambda u$$

Now:

$$\begin{aligned}(A - 5I)u &= Au - 5u \\ &= \lambda u - 5u \\ &= (\lambda - 5)u\end{aligned}$$

$\lambda - 5$ is an eigenvalue of $A - 5I$ with eigenvector u . If λ is an eigenvalue of A then $\lambda - 5$ is an eigenvalue of $A - 5I$. Therefore, the eigenvalues of $A - 5I$ are $-3, 0$.

Remark: If A is a square matrix of shape $n \times n$ and λ is an eigenvalue of A , then $\lambda - p$ is an eigenvalue of $A - pI$.

5. Question-5

The eigenvalues of a matrix A of shape 5×5 are $7, 5, 5, 5, -1$. If A is invertible, then find the eigenvalues of A^{-1} . What can you say about the eigenvectors of A^{-1} ?

If λ is an eigenvalue of A with eigenvector v , then:

$$\begin{aligned}Av &= \lambda v \\ A^{-1}Av &= \lambda A^{-1}v \\ Iv &= \lambda A^{-1}v \\ v &= \lambda A^{-1}v\end{aligned}$$

Rewrite this as:

$$\begin{aligned}\lambda A^{-1}v &= v \\ A^{-1}v &= \frac{1}{\lambda}v\end{aligned}$$

Provided $\lambda \neq 0$. Therefore, v is a eigenvector of A^{-1} with eigenvalue $\frac{1}{\lambda}$.

Remark: If A is an invertible matrix, then its eigenvalues are the reciprocals (inverses) of the eigenvalues of A and the corresponding eigenvectors are the same.

6. Question-6

A is a square matrix of shape 3×3 and has the eigenvalues $7, 0, 2$. What are the eigenvalues of A^3 ?

If λ is an eigenvalue of A with eigenvector v , then:

$$\begin{aligned}
Av &= \lambda v \\
A^2v &= \lambda Av \\
&= \lambda(\lambda v) \\
&= \lambda^2 v
\end{aligned}$$

Extending this idea, $A^k v = \lambda^k v$. Therefore, the eigenvalues of A^3 are $7^3, 0, 2^3$. In general, consider

$$B = (A^3 + 3A^2 - 2A + 5I)$$

If λ is an eigenvalue of A with eigenvector v , then:

$$Bv = (\lambda^3 + 3\lambda^2 - 2\lambda + 5)v$$

We see that $\lambda^3 + 3\lambda^2 - 2\lambda + 5$ is an eigenvalue of B with eigenvector v .

7. Question-7

If a square matrix A is singular (non-invertible), then it should certainly have eigenvalue λ . What is λ ?

A singular (non-invertible) matrix should always have zero as one of its eigenvalues. To see why this is true:

If A is invertible, then $Ax = 0 \implies x = 0$. If A is not invertible, then $Ax = 0$ has infinitely many solutions. In particular, there is some non-zero v such that $Av = 0 \implies Av = 0 \cdot v$. From this we see that v is an eigenvector of A with eigenvalue 0.

Remark: The eigenvalue 0 is intimately tied to the invertibility of a matrix.

8. Question-8

A is a 3×3 upper triangular matrix with diagonal entries 5, 4, 1. Is A

diagonalizable?

A looks like this:

$$A = \begin{bmatrix} 5 & \cdot & \cdot \\ 0 & 4 & \cdot \\ 0 & 0 & 1 \end{bmatrix}$$

The eigenvalues of an upper triangular matrix are its diagonal elements. If we expand along the first column, we get:

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 5 - \lambda & \cdot & \cdot \\ 0 & 4 - \lambda & \cdot \\ 0 & 0 & 1 - \lambda \end{vmatrix} \\ &= (5 - \lambda)(4 - \lambda)(1 - \lambda) \end{aligned}$$

A matrix A is diagonalizable if any of these two equivalent conditions are satisfied:

- if A is similar to a diagonal matrix.
- if there is a basis of eigenvectors of A for $\mathbb{R}^n$

One of the tests for diagonalizability is the following. If a matrix A has n distinct eigenvalues, then it is diagonalizable. The given matrix satisfies this test.

Question: If a triangular matrix has repeating diagonal entries (eigenvalues), is it still diagonalizable?

9. Question-9

Find the matrix A of shape 2×2 whose eigenvalues are 1, 4 and whose eigenvectors are $(3, 1)$ and $(2, 1)$ respectively.

A has 2 distinct eigenvalues, therefore, it is diagonalizable. A is similar to a diagonal matrix:

$$A = PDP^{-1}$$

The columns of P are the eigenvectors and the corresponding eigenvalues are in D .

$$P = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

We have P^{-1} :

$$P^{-1} = \frac{1}{3-2} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

We now have:

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

Remark: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $P^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

10. Question-10

Let $u, v \in \mathbb{R}^n$, with $u, v \neq 0$, and $A = uv^T$. Find all the eigenvalues and eigenvectors of A . Is A diagonalizable?

For example, if $u = (1, 2)$ and $v = (3, 4)$:

$$uv^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3 \ 4] = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

In general, uv^T is an $n \times n$ matrix. Consider Au :

$$\begin{aligned}
 Au &= (uv^T)u \\
 &= u(v^T u) \\
 &= (v^T u)u
 \end{aligned}$$

$v^T u$ is an eigenvalue for eigenvector u . Another eigenvalue is 0 and that would happen when the eigenvector is orthogonal to v . How do we establish that these are the only eigenvalues? Formally, let λ be some eigenvalue of A with eigenvector w . It is understood that $w \neq 0$:

$$\begin{aligned}
 Aw &= \lambda w \\
 uv^T w &= \lambda w \\
 (v^T w)u &= \lambda w
 \end{aligned}$$

We now consider two possibilities.

Case-1: w is orthogonal to v

$$\begin{aligned}
 (v^T w)u &= \lambda w \\
 0 &= \lambda w \\
 \lambda &= 0
 \end{aligned}$$

0 is one eigenvalue of A .

Case-2: w is not orthogonal to v

$$v^T w \neq 0$$

$$\begin{aligned}
 (v^T w)u &= \lambda w \\
 (v^T w)(v^T u) &= \lambda(v^T w) \\
 \lambda &= v^T u
 \end{aligned}$$

$v^T u$ is another eigenvalue of A .

A is diagonalizable. Why?

11. Question-11

If the eigenvalues of a 3×3 matrix A are 2, 2, 3, which of the following are certainly true?

- A is invertible
 - A is diagonalizable
 - A is not diagonalizable
-

A doesn't have any zero eigenvalues, hence A is invertible.

A may or may not be diagonalizable. We need to find the eigenvectors and see if we get three linearly independent eigenvectors.

12. Question-12

If $A^2 = I$, what could be the eigenvalues of A ?

Notice that since $AA = I$, A is invertible. Particularly, $A^{-1} = A$. If λ is an eigenvalue of A with eigenvector v , then $\frac{1}{\lambda}$ is an eigenvalue of $A^{-1} = A$ with eigenvector v :

$$\lambda = \frac{1}{\lambda} \implies \lambda^2 = 1 \implies \lambda = \pm 1$$

13. Question-13

The only eigenvectors of A are the non-zero elements of $\text{span}\{(1, 0, 0)\}$. Which of the following are true?

- A is not invertible
 - A has a repeated eigenvalue
 - A is not diagonalizable
-

A is not diagonalizable because it doesn't have three linearly independent eigenvectors that can span $\mathbb{R}^3$. Alternatively, $\mathbb{R}^3$ doesn't have a basis made up of eigenvectors of A .

What will be the first column of A ? Think about it.

Points (A) and (B) are not necessarily true Here is an example of a matrix that satisfies the conditions given in the question which is not invertible and doesn't have repeated eigenvalues.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} |A - \lambda I| &= (1 - \lambda) [(1 - \lambda)^2 + 3] \\ &= (1 - \lambda) (\lambda^2 - 2\lambda + 4) \end{aligned}$$

There is only real eigenvalue, namely, $\lambda = 1$. Any non-zero vector in $\text{span}\{(1, 0, 0)\}$ is an eigenvector of A with eigenvalue $\lambda = 1$.

14. Question-14

If $A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$, find A^{100} .

First we need to show that A is diagonalizable. This will allow us to express A as $A = PDP^{-1}$, then $A^{100} = PD^{100}P^{-1}$.

15. Question-15

If u is an eigenvector of A with eigenvalue λ_1 and v is an eigenvector of A^T with eigenvalue λ_2 , with $\lambda_1 \neq \lambda_2$, find $v^T u$.

We first write down what we know:

$$Au = \lambda_1 u$$

$$A^T v = \lambda_2 v$$

Multiply the first equation by v^T :

$$\begin{aligned}v^T(Au) &= \lambda_1(v^T u) \\(v^T A)u &= \lambda_1(v^T u) \\(A^T v)^T u &= \lambda_1(v^T u) \\(\lambda_2 v)^T u &= \lambda_1(v^T u) \\\lambda_2(v^T u) &= \lambda_1(v^T u) \\(\lambda_2 - \lambda_1)(v^T u) &= 0\end{aligned}$$

Since $\lambda_1 \neq \lambda_2$, $v^T u = 0$.

16. Question-16

Consider vectors $w_1 = (1, 1, 3, 5)$ and $w_2 = (-1, 1, 0, 0)$. Find the projection matrix, P , that projects vectors onto $S = \text{span}\{w_1, w_2\}$.

Normalize w_1 and w_2 :

$$\begin{aligned}v_1 &= \frac{1}{6}(1, 1, 3, 5) \\v_2 &= \frac{1}{\sqrt{2}}(-1, 1, 0, 0)\end{aligned}$$

$\{v_1, v_2\}$ is an orthonormal basis for S .

$$W = \begin{bmatrix} \frac{1}{6} & \frac{-1}{\sqrt{2}} \\ \frac{1}{6} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & 0 \\ \frac{5}{6} & 0 \end{bmatrix}$$

The claim is that $P = WW^T$ is a projection matrix for S . The full argument for why this is a projection matrix is given in this [document](#). Please ignore this for now as it is quite involved. You can take a look at it after the quiz, if you happen to be interested.

If you have just one column in W with unit norm, then $P = aa^T$. Since $\|a\| = 1$, the denominator seems to be missing, but is in fact $a^T a = 1$.

17. Question-17

If x and y are orthogonal and P is a projection matrix, what can you say about Px and Py ? Are they also orthogonal?

If P is a projection matrix corresponding to the vector a , then Px and Py are essentially scalar multiples of a , hence they can't be orthogonal. The conclusion is that Px and Py needn't be orthogonal. Thanks to Adithya Sabarish Saravanan for the argument.

18. Question-18

Find the point in the column space of $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ that is closest to $b = (1, 1, 0)$.

Solve for $\hat{x}$:

$$(A^T A)\hat{x} = A^T b$$

$\hat{x} \in \mathbb{R}^2$. The solution is $A\hat{x}$. To see why this is true, note that the point closest to b in the column space is the projection of b onto it. If the projection is $A\hat{x}$, then $b - A\hat{x}$ is orthogonal to the column space of A . This gives us:

$$\begin{aligned} A^T(b - A\hat{x}) &= 0 \\ \implies A^T A\hat{x} &= A^T b \end{aligned}$$

Therefore, $A\hat{x}$ is the required point in the column space.

19. Question-19

If A is a symmetric matrix, which of the following are true?

- Its column space is equal to its nullspace
- Its column space is equal to its row space
- Its column space is orthogonal to its nullspace

If $A^T = A$, then its columns and rows are the same, therefore, column space is equal to the row space. Since the row space is orthogonal to the nullspace, *in this case of a symmetric matrix*, the column space is also orthogonal to the nullspace. Note that this is not true in general.

20. Questions-20

Let A be an $m \times n$ matrix. If $Ax = b$ **always** has at least one solution for every $b \in \mathbb{R}^m$, how many solutions does $A^T y = 0$ have?

If b is in the column space of A for every $b \in \mathbb{R}^m$, then the column space of A has to be $\mathbb{R}^m$. That is, the columns have to span $\mathbb{R}^m$. If $A^T y = 0$, then y is orthogonal to all the columns of A (rows of A^T), which means that y is orthogonal to the column space of A , which is $\mathbb{R}^m$. The only vector in $\mathbb{R}^m$ that is orthogonal to all of $\mathbb{R}^m$ is the zero vector. Therefore, $A^T y = 0$ has exactly one solution, which is the trivial solution, $0 \in \mathbb{R}^m$.

21. References

These problems have been sourced from professor Gilbert Strang's textbooks: [Linear Algebra and its Applications](#).